

PROGRAMMABLE WEARABLES FOR HUMAN PERCEPTION

MULTIMODAL ACQUISITION FOR ROAD SAFETY

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CONTEXT AND STATE-OF-THE-ART

State of the Art (SOA)

- Infrastructure-based systems suffer from blind spots.
- Previous approaches (e.g., HoloLens) are intrusive and impractical for daily use.

Our Innovation

- Use of everyday wearable devices.
- Omi Glasses + Smart Bracelet for continuous sensing.
- Devices connected to a local smartphone for data processing and alerts.

REQUIREMENTS GATHERING

Prototyping and Technical Observation

- Practical exploration of the BrilliantSole SDK and firmware to map the existing communication workflow.
- Laboratory testing of sensors (Omi Glasses and Wristband) to identify hardware limitations such as latency and battery constraints.

Brainstorming and Technical Workshops

- Weekly meetings to discuss the feasibility of communication protocols (BLE vs. Wi-Fi).
- Definition of the sensor fusion logic and the need for timestamps to synchronize data.

State of the Art Analysis (Literature Review)

- Study of limitations in traditional infrastructure-based systems.
- Analysis of previous projects (e.g., SafeARCross).

ACTORS AND PERSONAS

Pedestrian (VRU):

- The human user.
Generates behavioural data and receives alerts.

Mobile App (Central Node):

- The local “brain”.
Aggregates, synchronizes, and makes decisions.

Infrastructure / Vehicle:

- Receivers of V2X warnings and transmitters of potential hazards.

USE CASE SCENARIOS



Primary Use Case: Imminent Collision Warning

1. Pedestrian approaches the road.
2. Wearables detect intention (head turns, walking speed decreases).
3. Mobile app processes the data and checks for an approaching vehicle.
4. App sends a haptic alert command (vibration) to the wristband.

USE CASE SCENARIOS



MICROPHONE (GLASSES)

_SIREN DETECTED
_MABULANCE APPROACHING



IMU HEAD (GLASSES)

_NO LATERAL ROTATION DETECTED
_THE GAZE DINDN´T CHECK THE SIDES



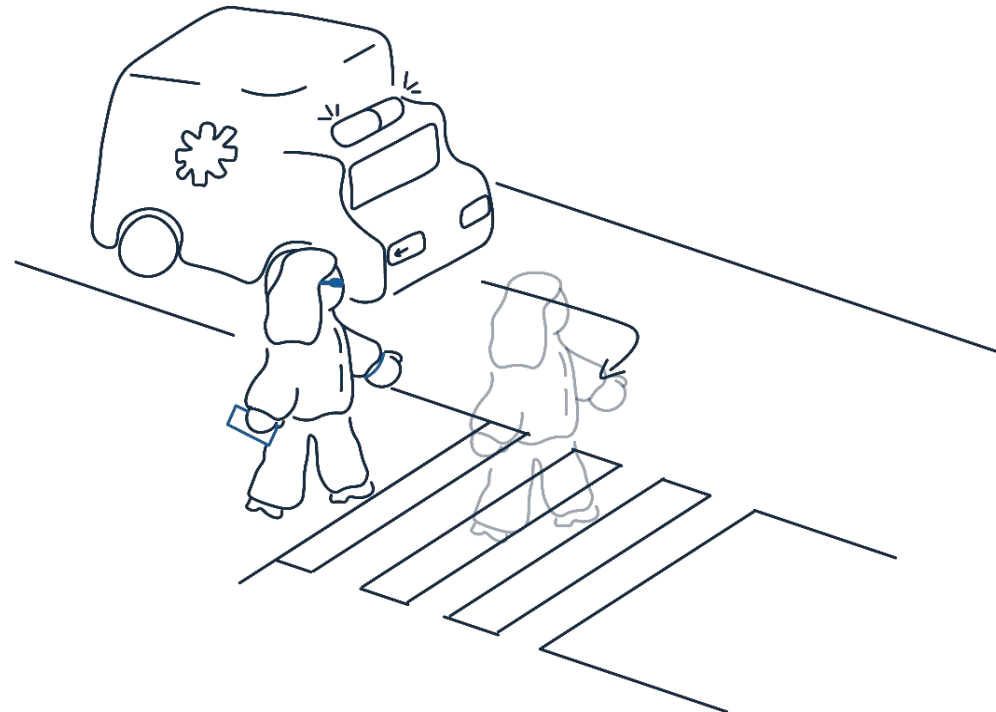
IMU WRIST/ WRISTBAND

_CONTINUOS MOTION
_4.8 KM/H
_NO SLOW DOWN



SMARTPHONE ACCELEROMETER

_ONGOING MOVEMENT
_WITHOUT STOPPING



TIMELINE UI



Primary Use Case: Imminent Collision Warning

1. Pedestrian approaches the road.
2. Wearables detect intention (head doesn't move, walking speed mantains).
3. Mobile app processes the data and checks for an approaching vehicle.
4. App sends a haptic alert command (vibration) to the wristband.

FUNCTIONAL REQUIREMENTS AND ELICITATION

What the system must do:

- Manage multiple independent connections simultaneously (multi-wearable support).
- Receive and interpret data streams (IMU, image, and audio).
- Send actuation commands (photo capture, vibration activation).

Our Innovation

- BrilliantSole SDK and firmware used as the basis for communication mapping and system development.

NON-FUNCTIONAL REQUIREMENTS

Performance (Communication):

- Evaluation in progress between BLE vs. Wi-Fi (trade-off: stability vs. throughput).

Data Integrity (Synchronization):

- Use of timestamps to align samples from multiple sensors (sensor fusion).

Autonomy and Customization:

- Need for custom firmware for the Omi Glasses (adjusting FPS and resolution).

SYSTEM ARCHITECTURE (BASELINE)

Sensor Layer:

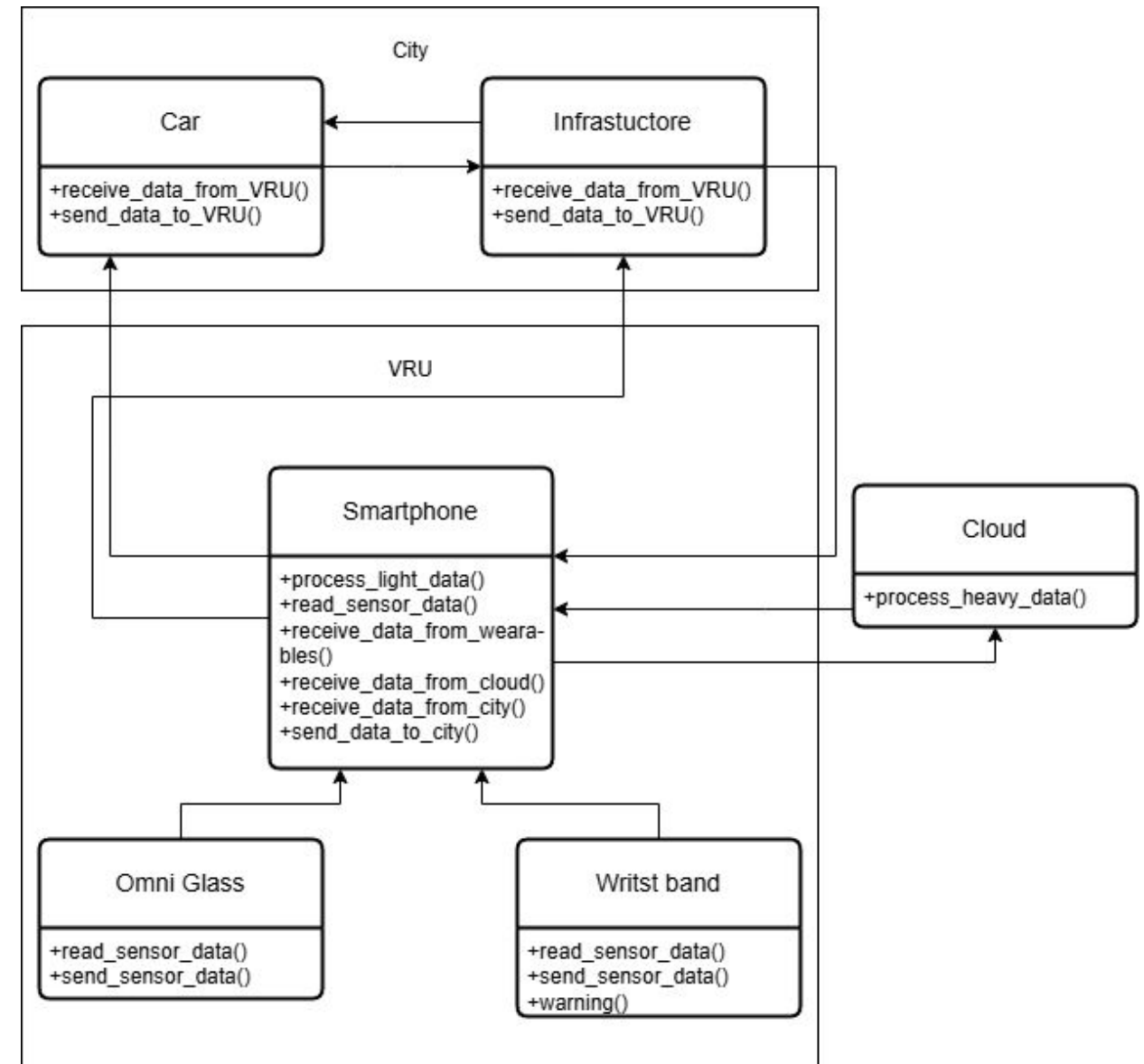
Omi Glasses & Wristband.

Aggregation Layer:

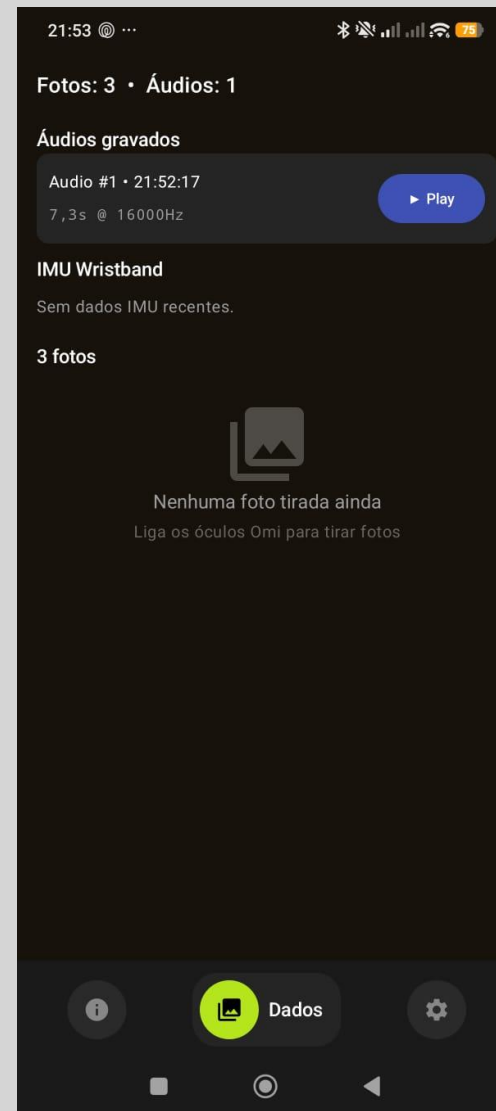
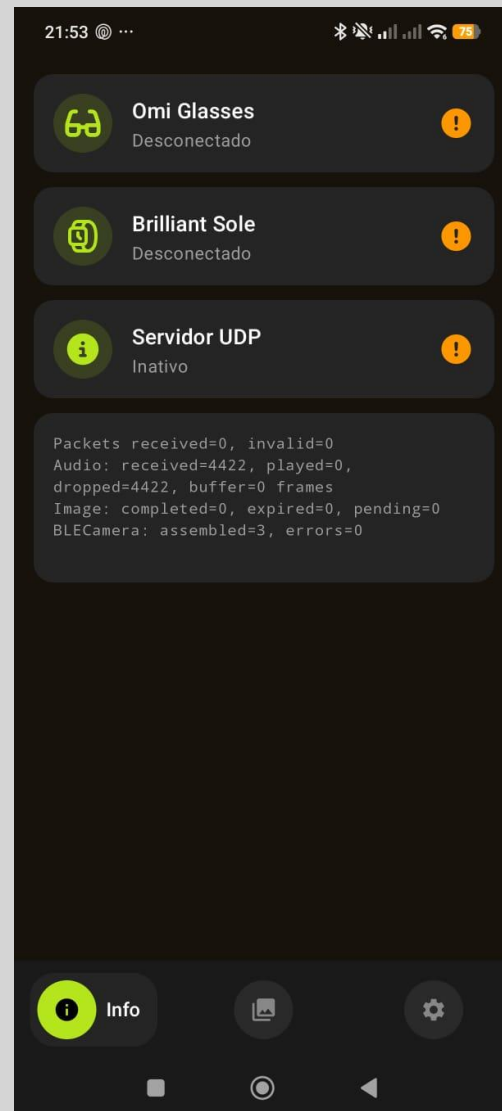
Kotlin App (management, synchronization, local ML).

Cloud / V2X Layer:

Communication with ATCLL and vehicles.



UI MOCK-UPS



DEMONSTRATION OF THE SKELETON (PROOF OF CONCEPT)

DEMO: MULTI-WEARABLE COMMUNICATION SKELETON



Thank You for Your Attention

Questions and Answers (Q&A)

